School of Computer Science
University of Nottingham

COMP3074 – Human-AI Interaction
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Information

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Late submission policy:	UoN default 5% per working day

(note marks cannot be released during exam time)

Overview

In this coursework you will be required to design and implement your own user-facing **interactive NLP-based system**, evaluate its performance and usability, write it up in a report, and make a demo video showing it in action. Using Python and a restricted selection of libraries, you will produce the following three (3) deliverables:

- **Deliverable 1**: an interactive NLP-based AI system, worth **30%** of the grade.
- **Deliverable 2**: a report of up to a **maximum of 2,000 words** (excluding citations, but including tables and figures) describing the architecture, implementation, and evaluation of the system, worth **50%** of the grade.
- **Deliverable 3**: a short video of **no more than 5 minutes** showing your system working, worth **20%** of the grade.

This document will describe each of those three deliverables, explain what they should contain, and how they will be evaluated.

Requirements & Specifications

Deliverable 1. Interactive NLP-based AI system

You should build a dialogue-based system which allows a user to complete a **transaction** (see below) with an AI, commonly known as a **chatbot**, or a **conversational AI**. The work will be based on what you have learned, using Python and a restricted set of allowed libraries[1]. A lot of Python libraries can automate much of the process, and you cannot get credit for work that you outsourced to a library. Any part of the assessment that is performed by a library will not be counted towards your grade.

The two key objectives for this coursework are:

- **Natural language processing architecture**: the system needs to include the architectural features that show what you have learned about NLP.
- **Conversational design**: the system should implement the **conversational design principles** as covered in lectures to ensure a usable and satisfactory user experience of your chatbot.

The system's main functionality should be focused on supporting the user in making a *transaction*, i.e., it should support users in completing a task, such as booking and ordering in a specific domain of your choice. Some examples (not an exhaustive list) of acceptable tasks include:

- Medical/clinical appointment reservation.

[1] Python standard library, numpy, scipy, scikit-learn (and dependencies), nltk **except for** nltk.chat

- Vehicle maintenance reservation.
- Personal care and salon appointment reservation.
- Sports court hire.
- Equipment hire.
- Repeat prescription renewal.
- Parcel dispatch.
- On-demand taxi dispatch.
- Public transport ticket purchase and renewal.
- Interactive fitness instruction.
- Step-by-step technical fault diagnosis.
- Guided assembly and maintenance tuition.
- Smart home environment control.
- Domestic maintenance and trade coordination
- Workplace desk and conference room administration.
- Academic module enrolment administration.
- Personal wardrobe curation.
- Your turn, be creative! If you are in doubt whether your idea satisfies the requirements, ask us in the lab.

1.1 Architecture

This section contains a non-exhaustive list of common tasks in interactive NLP-based systems (chatbots) that you are expected to implement. They are not all equal in complexity or importance, e.g., intent matching is necessary for any multi-usage chatbot while small talk is not.

Expected features:

1. Intent matching

Intent matching is a necessary part of any system which aims to do more than one task. It consists in taking in a user input and predicting the intent of the user, i.e., which function of the system the user is intending to use. Such functions could be storing/showing the user's name, outputting the current date and time, starting a transaction dialogue, or anything else described in this document. Intent detection can be done at different degrees of complexity, from simple text matching to similarity-based and machine learning-based approaches.

2. Identity management

Identity management refers to being able to acquire, memorise, and use the name of the person currently using the system. It consists in functions such as detecting name statements (e.g., "my name is X", "call me X", etc.) and storing the name to be used either in basic statements addressing the user, or in explicit name output functions (e.g., "what is my name?").

3. Transactions

A transactional dialogue system is a tool that allows for an artificial system to dialogue with a human operator **in order to achieve a specific set of goals**, which are **transactional** in nature, such as booking a train/plane ticket, a table in a restaurant, ordering an item from a store, etc.

Here is an example of a turn-by-turn dialogue between an operator and a travel booking system, to illustrate the range of tasks that such system would need to accomplish to properly function.

Turn	Author	Text

1	Bot	Hello, welcome to the Skynet Travel Agency, how may I help?
2	User	I would like to book a flight from London to Paris
3	Bot	Would that be a return trip or a single flight?
4	User	A return trip please
5	Bot	When were you thinking of going?
6	User	From the 10th to the 19th?
7	Bot	I'm sorry, I am going to need more details than that – please use the precise dates of your trip and whether you are flexible in your dates
8	User	10/11/2021 to 19/11/2021, and I am flexible
9	Bot	Thank you, I will search my database for available tickets. Please wait.
10	Bot	I have one flight available on the 10th from London to Paris, but no flight from Paris to London on the 19th, would you like to change the return date or book a single flight?
11	User	Change the date to the 20th

4. Information retrieval & question answering

A conversational information retrieval system is an information retrieval tool that interprets a *query* from a dialogue with a user and returns a set of one or more results which are deemed to be relevant, such as **direct answers**, **webpages**, **books**, **recipes**, **songs**, **movies**, etc. It requires having some form of database of documents and an indexing system. A question answering system can be seen as a special case of information retrieval where the documents are question-answer pairs, and only the answer is retrieved.

5. Small talk

A small talk function refers to the ability to have short discussions about shallow topics, such as "hello", "how are you", "how's the weather", etc. There is no expectation of long conversations due to them being one-off topics, but they can either be single turn (question, answer) or multi-turn (question, answer, and one or more follow-ups).

1.2 Conversational Design

Your chatbot should implement the following **conversational design principles** (as covered in lectures). You should also comment on the principles where appropriate in your report and video.

1. **Prompt design**: Have conversational markers and key phrases been used? Are the prompts clear and concise? Are prompts structured to support the user/task?
2. **Discoverability**: Is contextual help provided (e.g., what can the user do next?), and help to discover functionality and content?
3. **Error handling**: What error handling strategies are being used? Are they used effectively?
4. **Personalisation**: Is the experience somehow personalised to the user?
5. **Confirmation**: What confirmation strategies are used? Are these effective?
6. **Context**: Is context tracking used effectively?

*Deliverable 2. **Report***

You should write a report explaining and critically analysing your system, using the following structure:

1. **Title** – find a cool title for your system.
2. **AI statement** – explain your usage of generative AI (if any).
3. **Introduction** – introduce your system briefly.
4. **Chatbot Architecture** – technical description of your system. At least three points should be discussed: (a) **functionality** (what it does), (b) **implementation**: how you built it, and (c) **justification**: why you built it that way. Necessary images, flowcharts, and general illustrations[2] are welcome. Reference relevant parts of code in the report (if short), or in the project files (if long).
5. **Conversational Design** – describe how you implemented the Conversational Principles.
6. **Evaluation** – how you evaluated your project, i.e., **usability testing** and **performance testing**.
7. **Discussion** – Reflect on the **results of your evaluation** and **your chatbot's application area**.

You are strongly encouraged to follow this structure if you are not an experienced writer of scientific reports (and you are not). The report should me a maximum of **2,000 words** (+/-10%). We recommend that the sections on Chatbot Architecture and Conversation Design together should have approximately 2/3 of the overall word count.

There is no strict minimum number of words, but from past experience, if your report is very short, it is likely that you forgot something. We expect the maximum word count to be one key challenge of this coursework.

Use the single column ACM template available for Microsoft Word, LaTeX (use `sample-manuscript.tex` for submissions), and Overleaf. **We expect you to put some effort in the presentation of your coursework**.

Your submission should be a single PDF document.

Deliverable 3. ***Demonstration video***

You should produce a short, narrated demonstration video of **up to five minutes (+/-10%)**, that demonstrates your system being used. This can be achieved by screen capture of you using the chatbot live. The video should have some form of audio narration of you explaining what the chatbot is doing. In the video, you should demonstrate the different features you implemented, and the conversational design principles you followed.

It is not sufficient to show/narrate your code in a text editor or using slides to explain your chatbot. Show the system as you use it, with you explaining what is happening in your own words. Make sure the audio of the video is understandable and not sped up to the point of unintelligibility. Make sure you

are covering all the features you implemented.

Getting Started

Make good use of labs and lectures to ask questions related to the coursework. There will also be a number of coursework-focused lab sessions during which you can ask for specific help. Finally, a good starting point would be to ensure you satisfy the requirements of the intermediate formative checkpoint.

[2] https://www.diagrams.net is a great tool to make all sorts of charts.

Submission Instructions

Submit all 3 deliverables on Moodle:

- Deliverable 1 should be **a zipped file containing your project**.
- Deliverable 2 should be **a PDF document containing your report**.
- Deliverable 3 should be **a video in a sensible format** readable on university machines.

Assessment Criteria

The coursework is assessed across seven criteria. Each criterion is assigned to the highest band for which the work demonstrates the required standard; each band incorporates the core requirements of all lower bands. The final mark is determined by the lowest band achieved across all criteria of that component (system, report, or video), as stronger performance in one area cannot compensate for a lower standard elsewhere. For example, a profile of multiple criteria in the 70–79 band and one in the 50–59 band results in a final band of 50–59.

Criteria will be marked based on the following band standards:

Band	Standard
0-39 Fail	A serious deficiency or absence of evidence prevents an essential outcome from being demonstrated.
40-49 Basic pass	Essential outcomes are demonstrated, but substantial weaknesses may remain.
50-59 Competent	Main requirements are met with basic explanation and evidence, but depth or robustness is limited.
60-69 Good	Coherent, well-supported work. Shows sound judgement and effective treatment of common challenges.
70-79 Excellent	Sustained critical judgement, depth and convincing evidence are present across the criteria.
80-100 Outstanding	Exemplary achievement is sustained throughout, with distinctive insight or exceptionally effective solutions.

	Criterion	Assessment Indicators
Report	**Description of architecture**	• Depth and detail of the description of the architecture and implementation • Use of adequate examples and illustrations
	Description of conversational design principles	• Detail of the description of the conversational designed and how they were implemented • Use of adequate examples and illustrations
	Evaluation	• Breadth and depth of the evaluation • Rigorous and reproducible evaluation • Rigorous performance testing • Rigorous usability testing • Insightful interpretation of the results
	Discussion	• Depth and detail of the discussion of the results • Insightful contextualisation of the work with respect to the intended application area
	Quality of writing and presentation	• Adherence to the word limit and template • Good use of citations and bibliography • Professional formatting of figures and tables • Effective structure and clear writing
System	**Core architecture**	• Execution of the code works without error • Inclusion of necessary data files or correct paths • Presence of comments to explain the logic • Features are well implemented and robust
	Implementation of conversational design principles	• Coverage of conversational design principles • Quality of implementation • Robustness of implementation
Video	**Video submission**	• Concision and adherence to the time constraint. • Structure and pace of the video • Clarity (audio, visual,
	Demonstration of features	• Coverage of all the features • Accuracy and relevance of explanations • Use of examples and illustrations to illustrate features and limitations
	Demonstration of conversational design principles	• Coverage of all the conversational design principles • Accuracy and relevance of explanations • Use of examples and illustrations to illustrate features and limitations

Academic Integrity

Generative AI, mostly in the form of large language models, have become embedded in several applications and increasingly hard to avoid. For that reason, we try to make our generative AI policy as clear as possible:

What is allowed:	What is not allowed
Where it's unavoidable (e.g., embedded in search engine). As part of assisted coding (code completion, bug finding). To help you understand complex topics.	To write the report in your place (even if you rephrase it later on) To write the code for you (even if you reshape it afterwards).

Our plagiarism policy is simple: **do not plagiarise**. Plagiarism cases (whether based on AI, other work, or your own, uncited prior work) will be sent to the academic misconduct committee. To avoid accidental plagiarism:

- Cite the work you're getting your inspiration from using a consistent referencing system.
- Reference the code you're using. Preferably in the comment where you're using it.